草稿

State

$$\nu_t = \frac{1}{t}\sum_{j=1}^{t} v_j \tag{1}$$

$$\kappa_t = \frac{1}{t}\sum_{j=1}^{t} k_j \tag{2}$$

$$D_t \triangleq \sum_{j=1}^{t}(v_j - \nu_t)(k_j - \kappa_t)^\top \tag{3}$$

Recurrence

$$\nu_t = \nu_{t-1} + \frac{1}{t}(v_t - \nu_{t-1}) \tag{4}$$

$$\kappa_t = \kappa_{t-1} + \frac{1}{t}(k_t - \kappa_{t-1}) \tag{5}$$

$$D_t = D_{t-1} + \frac{t-1}{t}(v_t - \nu_{t-1})(k_t - \kappa_{t-1})^\top \tag{6}$$

Sequence Parallel

$$P_t = \mathrm{PrefixSum}(v)_t \tag{7}$$

$$R_t = \mathrm{PrefixSum}(k)_t \tag{8}$$

$$M_t = \mathrm{PrefixSum}(vk^\top)_t \tag{9}$$

$$D_t = M_t - \frac{1}{t}P_t R_t^\top \tag{10}$$

Output

$$\begin{cases} \Delta q_t = q_t - q_{t-1} & (11) \\ \beta_t = \dfrac{\exp(k_t^\top q_t)}{(t-1) + R_{t-1}^\top q_t} & (12) \\ o_t = (1-\beta_t)(o_{t-1} + (i-1)^{-1} D_t \Delta q_t) + \beta_t v_t & (13) \end{cases}$$